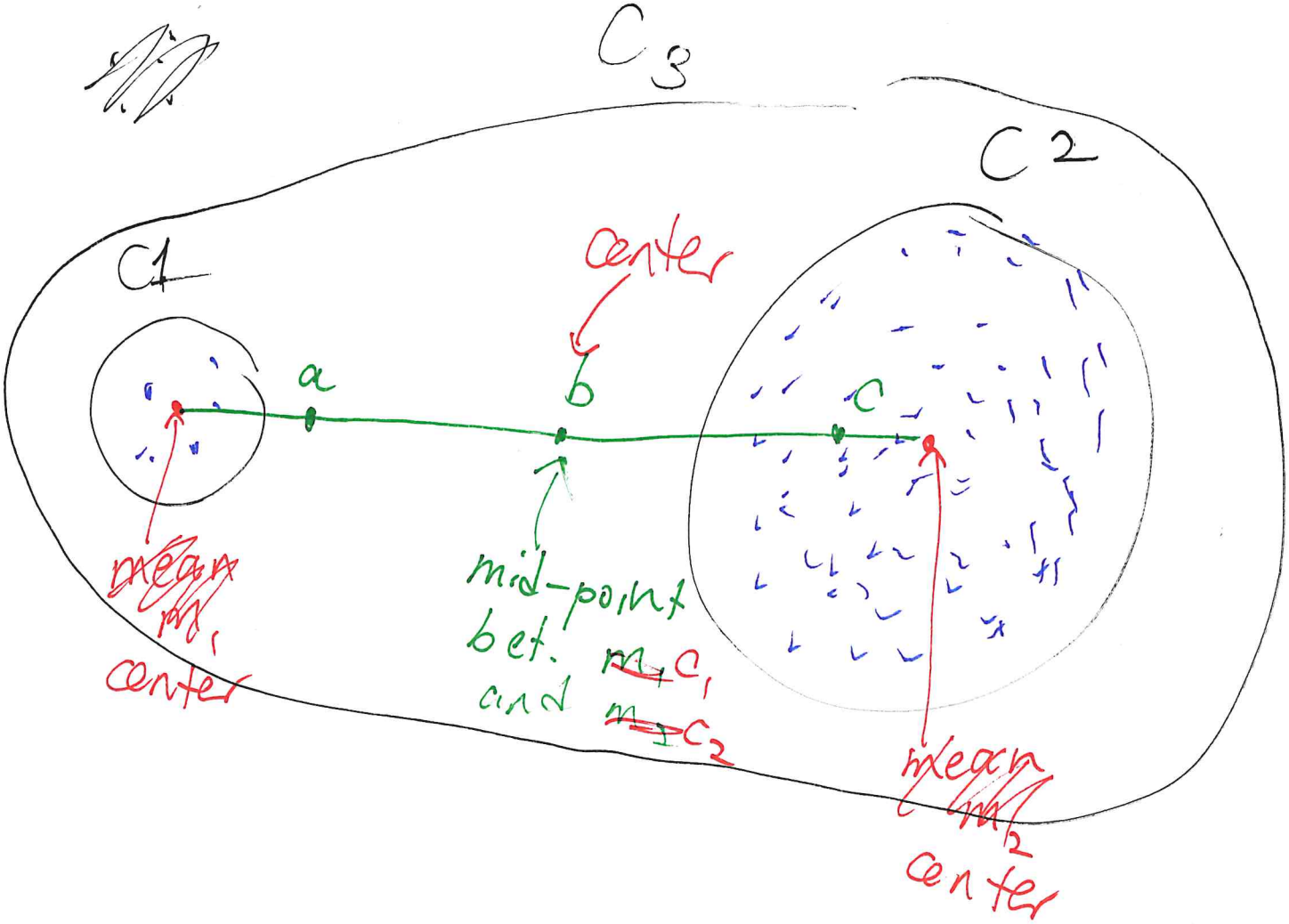
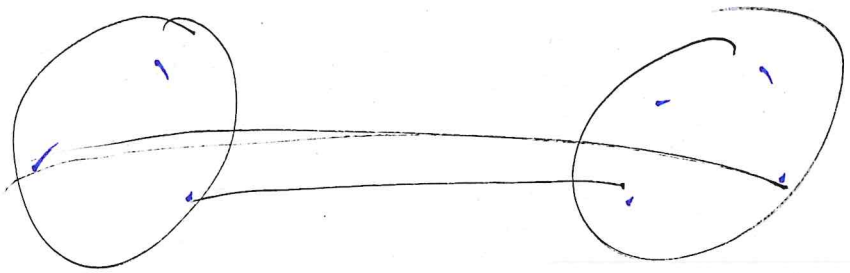
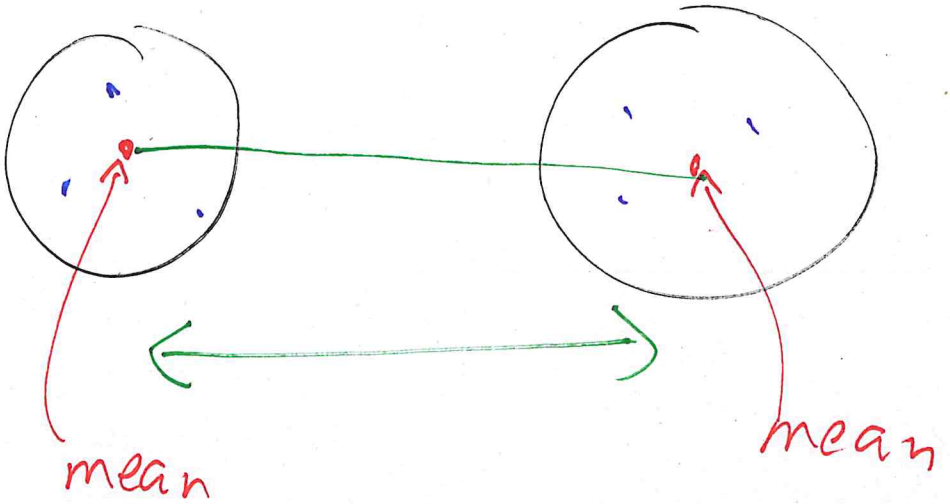
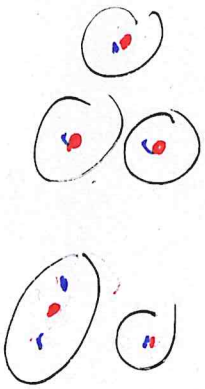
COMP5331 (23 Sept)

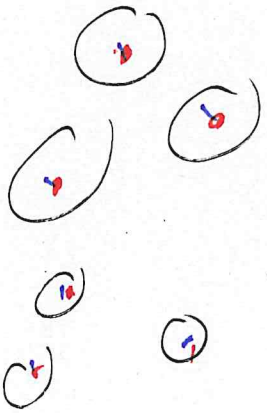
minority matters

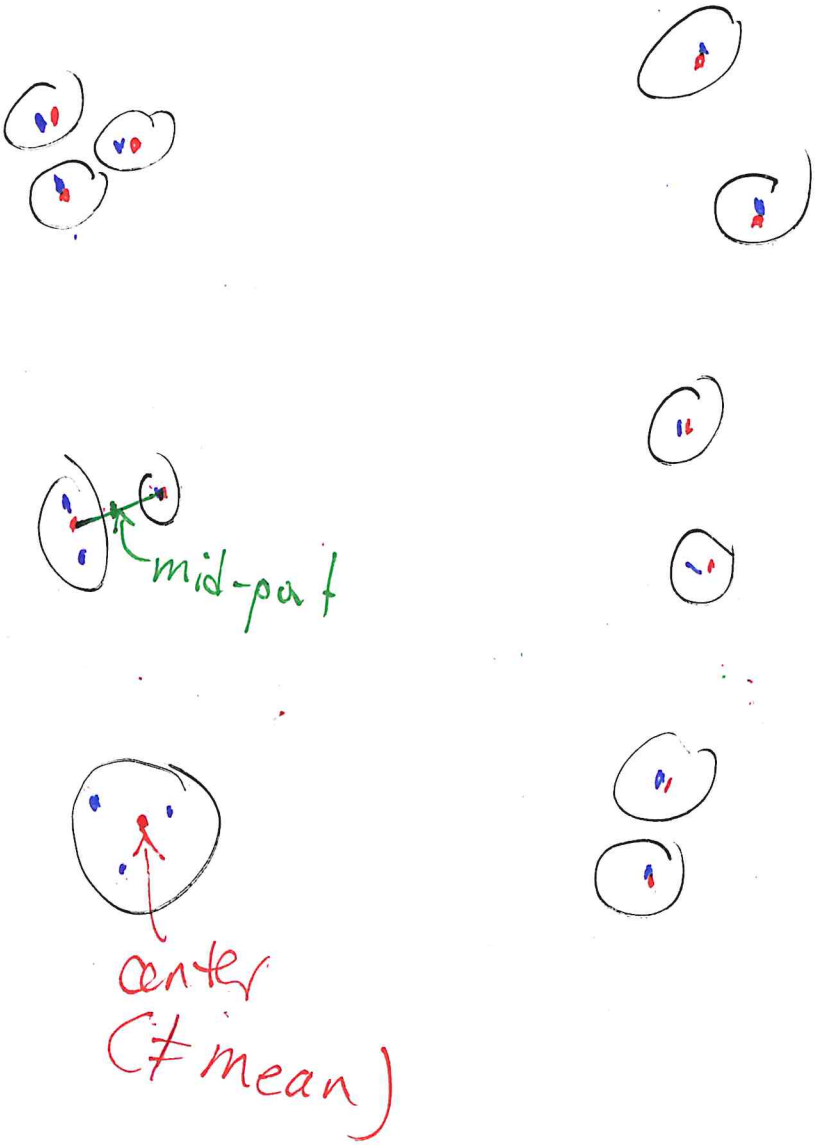





mean
mean






mid-pt
center
(≠ mean)

COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Exercise 3
Clustering

Q1 Agglomerative Approach

There are four two-dimensional points:

x_1 (1, 2),
x_2 (2, 2),
x_3 (5, 2),
x_4 (6, 1).

(a) Assume that we adopt the Euclidean distance metric as the distance between any two points. Please write a matrix where its entries correspond to the pairwise distances between any two points.
(b) Please use the agglomerative approach to group these points with distance median linkage. Draw the corresponding dendrogram for the clustering. You are required to specify the distance metric in the dendrogram.

Q2 Divisive Approach

There are the following four records with three binary attributes, namely A, B and C.

	A	B	C
1	1	1	0
2	1	1	0
3	1	0	1
4	0	0	1

Please use monothetic approach to perform hierarchical clustering over these records. If there is any tie for choosing the attributes, we choose the attributes in the order: A, B and C.
Draw the dendrogram (without specifying the distance metric).

Ex. 3.

Q1 (a)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	1	0		
x_3	4	3	0	
x_4	5.10	4.12	1.41	0

$\sqrt{(1-2)^2 + (2-2)^2}$

(b)

	x_1	x_2	x_3	x_4
x_1	0			
x_2	1	0		
x_3	4	3	0	
x_4	5.10	4.12	1.41	0

center : $(x_1) : (1, 2)$
$(x_2) : (2, 2)$
$(x_3) : (5, 2)$
$(x_4) : (6, 1)$

$\Rightarrow$

	$(x_1 x_2)$	x_3	x_4
(x_1, x_2)	0		
x_3	3.5	0	
x_4	4.61	1.41	0

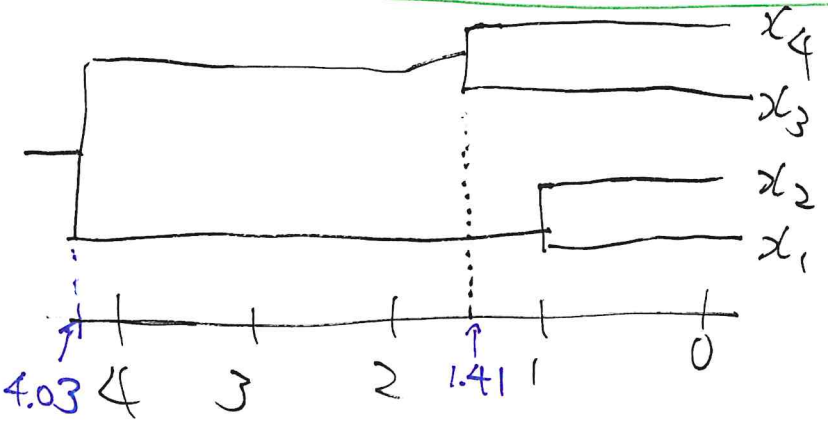
center : $(x_1 x_2) : (1.5, 2)$
$(x_3) : (5, 2)$
$(x_4) : (6, 1)$

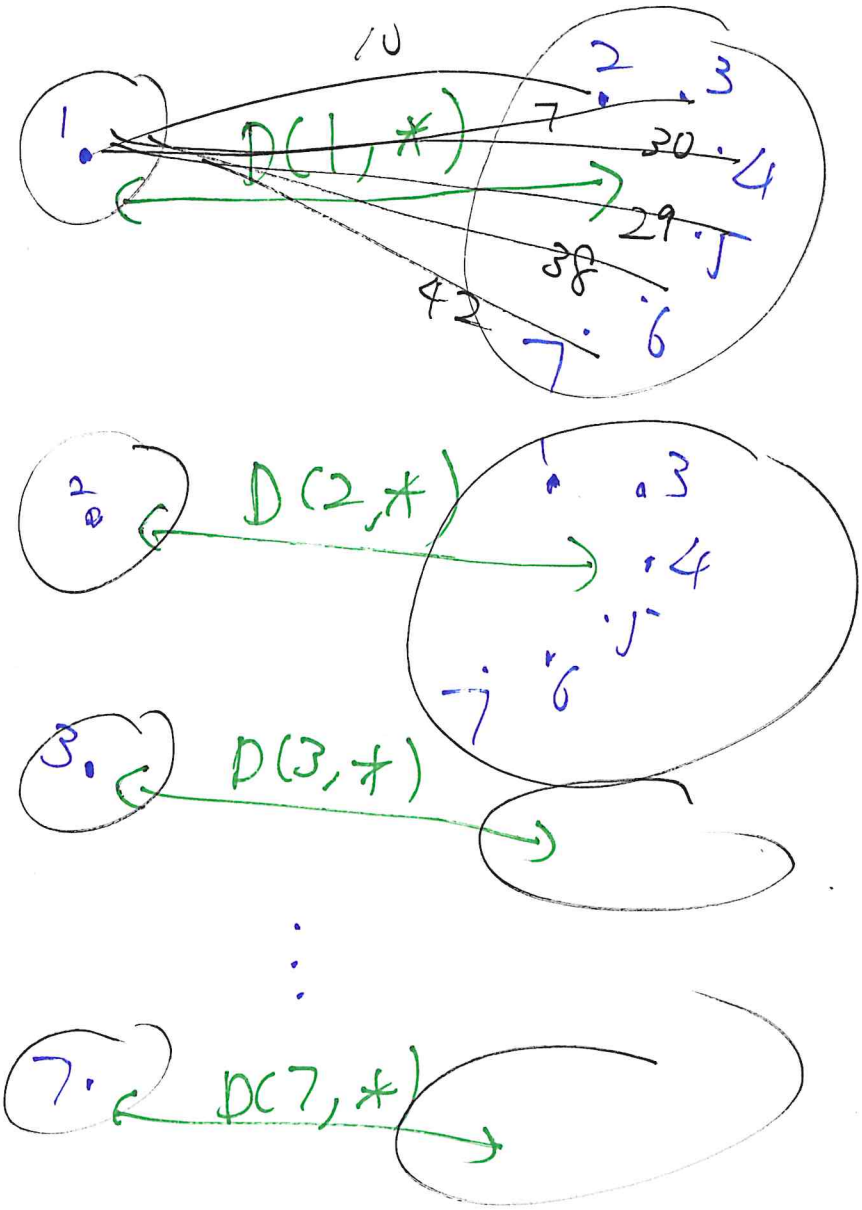
~~1~~

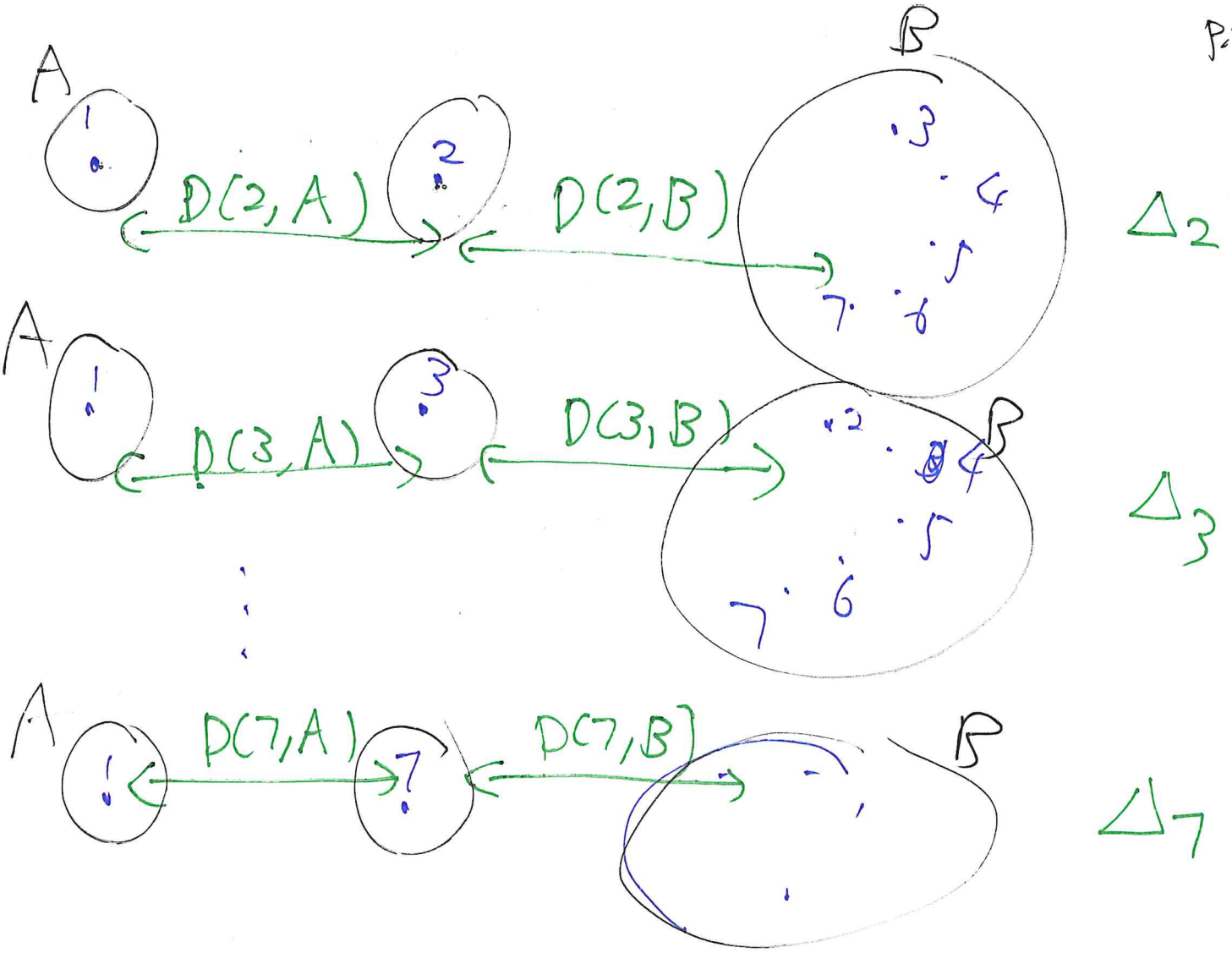
$\Rightarrow$

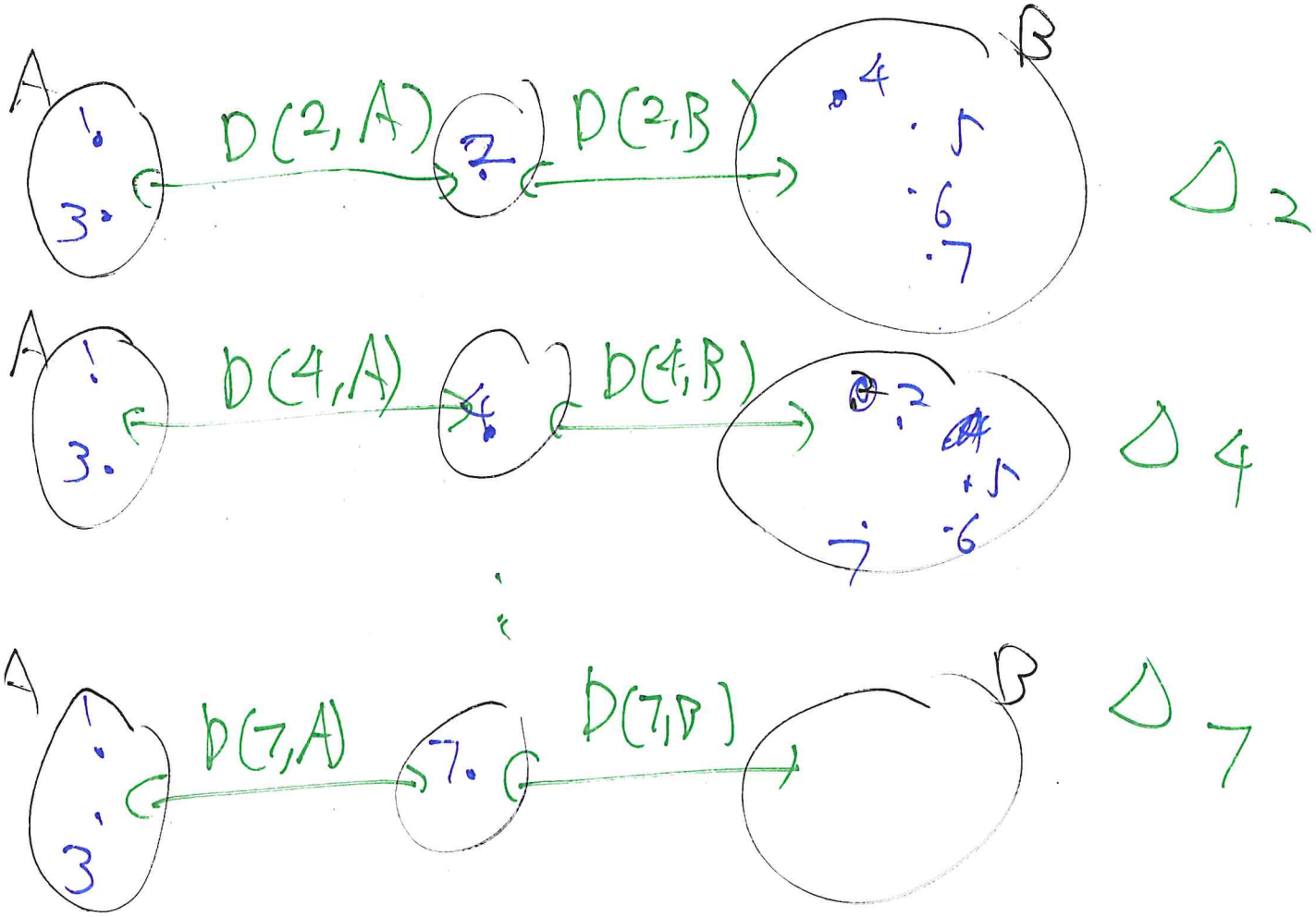
	$(x_1 x_2)$	$(x_3 x_4)$
(x_1, x_2)	0	
$(x_3 x_4)$	4.03	0

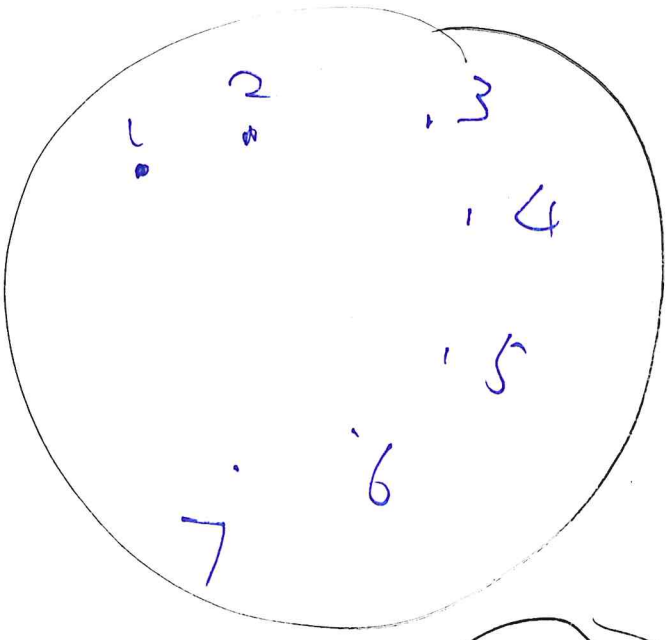
center : $(x_1, x_2) : (1.5, 2)$
$(x_3, x_4) : (5.5, 1.5)$

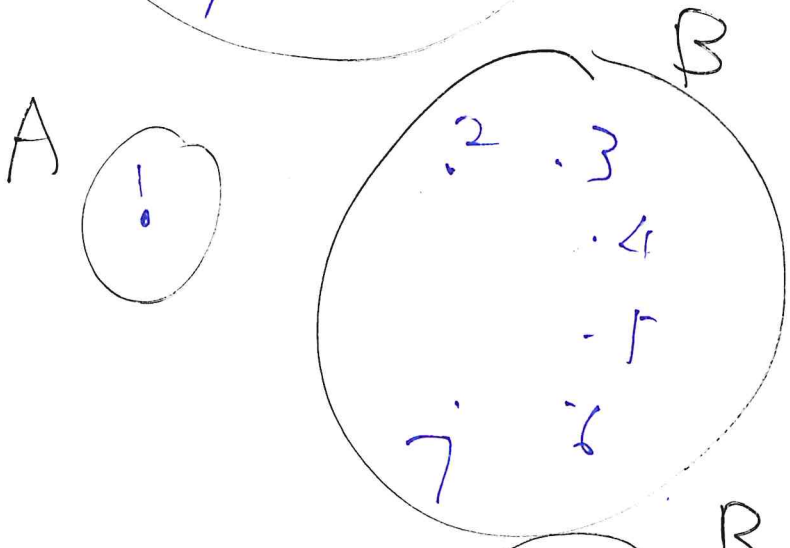


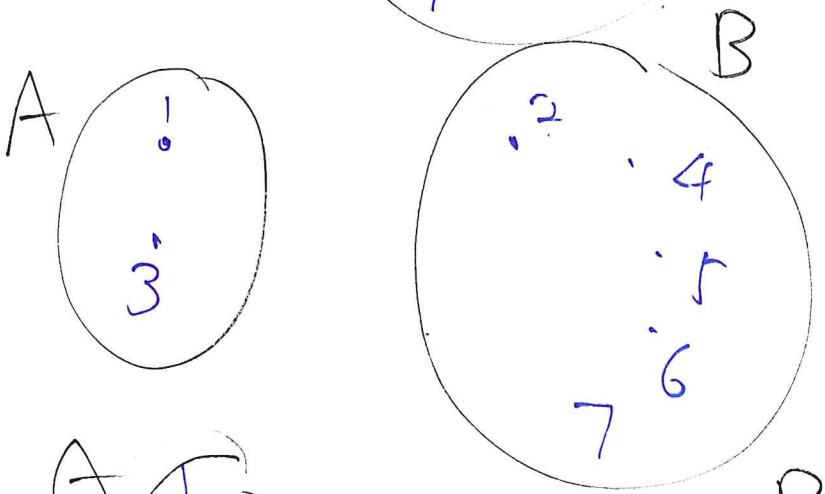

10
1
2
3
7
30
4
29
5
38
42
7
6
D(1,*)
2
D(2,*)
3
4
5
6
7
3
D(3,*)
7
D(7,*)

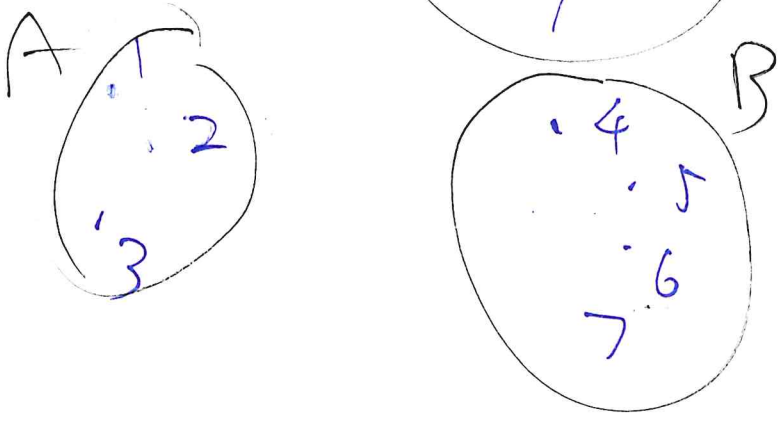

A
B
1
2
3
4
5
6
7
D(2,A)
D(2,B)
Δ_2
D(3,A)
D(3,B)
Δ_3
D(7,A)
D(7,B)
Δ_7


A
B
1
3
D(2,A)
2
D(2,B)
4
5
6
7
Δ 2
A
1
3
D(4,A)
4
D(4,B)
2
5
6
7
Δ 4
A
1
3
D(7,A)
7
D(7,B)
B
Δ 7


1
2
3
4
5
6
7


A
1
B
2
3
4
5
6
7


A
1
3
B
2
4
5
6
7

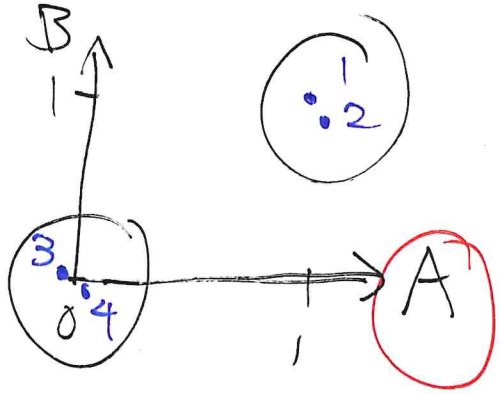

A
1
2
3
B
4
5
6
7

correlation bet. two attributes

e.g. 1

A and B are very correlated

A	B
1	1
1	1
0	0
0	0

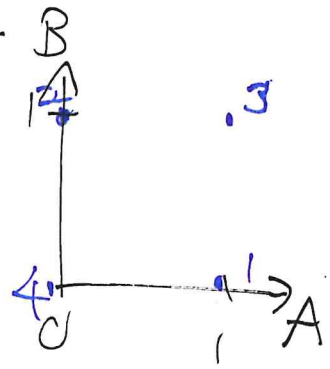


$\chi^2_{AB} = 4$

e.g. 2

A and B are independent

A	B
1	0
0	1
1	1
0	0



$\chi^2_{AB} = 0$

if χ^2_{AB} is larger,

A and B are more correlated

A:

avg. correlation bet. A and another attribute

Ex.3

Q2

B \ A	1	0
1	2	0
0	1	1

C \ A	1	0
1	1	1
0	2	0

C \ B	1	0
1	0	2
0	2	0

$\chi^2_{AB} = 1.33$ $\chi^2_{AC} = 1.33$ $\chi^2_{BC} = 4$

$$\chi^2_{AB} = \frac{(2\times 1 - 1\times 0)^2 \times 4}{(2+0)(2+1)(1+1)(1+0)} = 1.33$$

For attribute A,

$\chi^2_{AB} + \chi^2_{AC} = 2.66$

For attribute B,

$\chi^2_{AB} + \chi^2_{BC} = 5.33$ ← choose B

For attribute C,

$\chi^2_{AC} + \chi^2_{BC} = 5.33$

We divide the data into 2 groups.

{1, 2) and (3, 4)

Dendrogram

